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Subject : DSA

Unit 2, S14, SLO1 - Circular Doubly Linked Lists – Deletion

Algorithm to delete the first node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 8

[END OF IF]

Step 2: SET PTR = START

Step 3: Repeat Step 4 while _____ != START

Step 4: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 5: SET PTR -> NEXT = _____

Step 6: SET _____ = PTR

Step 7: FREE _____

Step 8: SET START = _____

Answer: Step 3: while PTR -> NEXT != START

Step 4: SET PTR = PTR -> NEXT

Step 5: SET PTR -> NEXT = START -> NEXT

Step 6: SET START -> NEXT -> PREV = PTR

Step 7: FREE START

Step 8: SET START = PTR -> NEXT

Algorithm to delete the last node

Step 1: IF START = NULL

Write UNDERFLOW

Go to Step 8

[END OF IF]

Step 2: SET PTR = START

Step 3: Repeat Step 4 while PTR -> NEXT != _____

Step 4: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 5: SET _____ = START

Step 6: SET _____ = PTR -> PREV

Step 7: FREE _____

Step 8: EXIT

Answer: Step 3: while PTR -> NEXT != START

Step 4: SET PTR = PTR -> NEXT

Step 5: SET PTR -> PREV -> NEXT = START

Step 6: SET START -> PREV = PTR -> PREV

Step 7: FREE PTR

Step 8: EXIT